SUPERVIOR'S RECOMMENDATION

LETTER OF APPROVAL

ACKNOWLEDGEMENT

First of all we would like to express our sincere thanks to the Department of Humanities And Social Science for providing us with the opportunity to explore our interest and ideas in the software development field through this project.

We are also very grateful to our project supervisor, **Avinash Maskey** sir who gave us his valuable time, suggestion, help, support, supervision and guidance to solve different problems that we have faced during our project. Without whom our project wouldn't be updated to this point.

In the end, we would also like to thank the Tribhuvan University for giving us this opportunity to help us understand the project ethics at this early stage and help us to evaluate our knowledge and expand it a little more.

ABSTRACT

A decade ago, only 10 percent of Nepalese were connected to the internet. Now, almost the entire population has access to the network, according to the Nepal Telecommunications Authority. Buyers miss out on the convenience of shopping online, merchants have no online marketplace to sell goods, and lower-skilled people cannot gain employment in delivery services, which can be a route out of poverty. E-commerce is among the few sectors with the potential to add value, helping these countries to diversify their economies and graduate from the LDC (Least Developed Country) group. This proposal identifies the current use of incremental e-business strategies to build and develop a reliable website. The proposal aims to provide mechanisms to positively influence each stage of the buying cycle, ensuring that customers return to ToyStore online repeatedly. To implement the ToyStore website, the approach involves setting up an online e-commerce store with user-friendly features. Then, gradually improve the customer experience in stages, and finally implement Direct Online Sales between business and consumer (B2C) through incremental integration of electronic payment methods. The new ToyStore website will include blogs, straightforward language, high-quality pictures, and sales prediction (using linear regression) for the seller to plan accordingly.

Key Words: Online purchase, B2C, Ecommerce, Customer awareness, Storyboard, Online marketplace, Digital inclusion, Poverty alleviation and Linear regression.

TABLE OF CONTENTS

SUPERVIOR'S RECOMMENDATION ..... **i**
LETTER OF APPROVAL ..... **ii**
ACKNOWLEDGEMENT ..... **iii**
ABSTRACT ..... **iv**
LIST OF FIGURES ..... **vii**
LIST OF TABLES ..... **viii**
LIST OF ABBREVIATIONS ..... **ix**
CHAPTER 1: **10**
INTRODUCTION **10**
1.1 Introduction 10
1.2 Problem Statement 11
1.3 Objectives 11
1.4 Scope and Limitations 11
1.4.1 Scope 11
1.4.2 Limitations 12
1.5 Report Organization 12
CHAPTER 2: **13**
BACKGROUND AND LITERATURE REVIEW **13**
2.1 Literature Review 13
CHAPTER 3: System Analysis and Design **14**
3.1 System Analysis 14
3.1.1 Requirement Analysis 15
3.1.2 Feasibility Analysis 18
3.1.3 ER-diagram ..... **Error! Bookmark not defined.**
3.1.4 Process Modeling (DFD) 20
3.2 System Design 22
3.2.1 Architectural Design 22
3.2.2 System Flowchart 25
3.2.3 Database Schema Design 28
3.2.4 Interface Design 29
CHAPTER 4: **31**
IMPLEMENTATION AND TESTING **31**

4.1 Implementation 31

4.1.1 Utilized Tools (CASE tools, Programming language, Database platforms) 31

4.1.2 Implementation Details of Modules (Description of procedures/functions) 32

4.2 Testing 34

4.2.1 Test Cases for Unit Testing 34

Table 4.2.1: User Sign in/Sign up Test Case **34**

4.2.2 Test Case for System Testing 35

Table 4.2.2: System Test Case **35**

4.2.3 Test Case for Admin 37

Table 4.2.3: Admin Dashboard Test Case **37**

CHAPTER 5: **40**

CONCLUSION AND FUTURE RECOMMENDATIONS **40**

5.1 Knowledge Gained / Results 40

5.2 Conclusion 41

5.3 Future Recommendations 41

REFERENCES **42**

LIST OF FIGURES

Figure 3.1 Incremental Methodology 14
Figure 3.2 Use Case Diagram 17
Figure 3.3 Gantt Chart 19
Figure 3.4 ER-Diagram of Ecommerce website ..... **Error! Bookmark not defined.**
Figure 3.5 0 Level DFD Ecommerce Website 20
Figure 3.6 Level 1 DFD Ecommerce Website 21
Figure 3.7 System Architectural design 24
Figure 3.8 Admin Flowchart 25
Figure 3.9 User Flow Chart 27
Figure 3.10 Database Schema 29
Figure 3.11 Admin Dashboard UI 30
Figure 3.12 User Dashboard UI 30

LIST OF TABLES

Table 3.1 Gantt chart table……11
Table 4.2.1: Login Page Test Case 34
Table 4.2.2: System Test Case..... 35
Table 4.2.3: Admin Dashboard Test Case 37

LIST OF ABBREVIATIONS

API	Application Programming Interface.
CDN	Content Delivery Network.
B2C	Business to Consumer.
DB	Database.
SSL	Secure Sockets Layer.
HTTPS	Hypertext Transfer Protocol Secure.
CMS	Content Management System.
SEO	Search Engine Optimization.
UI	User Interface.
UX	User Experience.
JWT	JSON Web Token.
IDE	Integrated Development Environment.
ORM	Object-Relational Mapping.

CHAPTER 1:

INTRODUCTION

1.1 Introduction

The landscape of e-commerce for toys and sports in Nepal has undergone a transformative evolution in the past decade. A decade ago, access to the internet was limited, with only 10 percent of the Nepalese population connected. However, recent advances in digital connectivity, facilitated by the Nepal Telecommunications Authority and other private entities, have led to a remarkable shift. The increased internet penetration has connected nearly the entire population and paved the way for the growth of online marketplaces. This shift has had profound implications for the toy and sports industry in Nepal.

Previously, buyers missed out on the convenience of online shopping, and merchants lacked a dedicated online marketplace to showcase and sell their products. Additionally, the potential for employment in delivery services, particularly for those with lower skills seeking an avenue out of poverty, was largely untapped. E-commerce, especially in toys and sports, emerges as a promising sector capable of adding substantial value to the economy. To fully leverage this growth, integrating advanced tools like price and sales prediction models is crucial. By analyzing historical price and sales data, merchants can forecast trends, optimize pricing strategies, and make data-driven decisions to enhance their competitiveness. Predictive analytics can help anticipate demand fluctuations, adjust inventory levels accordingly, and implement targeted promotions to boost sales. These insights can also assist in better understanding consumer behavior and preferences, ultimately improving customer engagement and satisfaction.

This transformation holds the potential to diversify the economic landscape of Nepal and contribute to the country's graduation from the category of Least Developed Countries (LDCs). The ongoing digital revolution presents a unique opportunity to harness e-commerce for toys and sports, fostering economic growth, digital inclusion, and improved access to recreational and sporting goods for the Nepalese population.

1.2 Problem Statement

Here are some of the problem statements we discovered:

- Need to enhance customer awareness and engagement due to a gap in educating potential buyers as they might be skeptical and only prefer face to face transactions.
- Building trust in online transactions remains a challenge.
- Absence of well-established e-commerce infrastructure hinders merchants' effectiveness.
- Providing a seamless and user-friendly online experience is essential for smooth transition from traditional retail to e-commerce.
- Security and privacy of online transactions are critical concerns requiring robust solutions.
- Logistical challenges in delivery and return as ensuring in timely deliveries and easy returns, is crucial to meet customer expectations in e-commerce.

Addressing these issues is crucial for unlocking the full potential of e-commerce, fostering economic development, and improving the online retail landscape in Nepal.

1.3 Objectives

The objectives of our platform are:

- To increase customer awareness and engagement through targeted marketing and educational initiatives.
- To develop a dedicated e-commerce infrastructure tailored to the toy and sports industry in Nepal.
- To implement robust security measures to address privacy concerns and ensure secure online transactions.
- To optimize user experience and facilitate a smooth transition from traditional retail to e-commerce for toys and sports.
- To equip the Admin with data for strategic planning, utilizing sales forecasting and predictive analytics.

1.4 Scope and Limitations

1.4.1 Scope

The scope of the Ecommerce Website is to provide a comprehensive and user-friendly platform tailored specifically for sport goods. At its core, the system's functionalities

encompass efficient transactions, ease of update for admins to products, reduce burden of tension, and get information about the products you want, allow users to have secure transactions. It aspires to bring efficiency, transparency, and collaboration to the forefront of ecommerce activities.

1.4.2 Limitations

- As it is a web-based application it needs constant internet connection.
- System effectiveness relies on precise data entry, with errors potentially leading to skewed insights.
- Digital Literacy
- Financial Constraints

1.5 Report Organization

Introduction

In this part, we talk about the start of the system, what it aims to do, and its limitations.

Background Study and Literature Review

Here, we look at the research done in this field before and talk about other similar systems that already exist for Ecommerce Website which sells Products.

System Analysis and Design

This section is about figuring out what the system needs to do. We look at how it should work, what it should have, and check if it's possible to make. We also draw different diagrams to show how it all fits together.

Implementation and Testing

Now, we get into making the system real. We talk about the tools we use to build it, how we actually make it work, and how we test to make sure it's doing what it should.

CHAPTER 2:
BACKGROUND AND LITERATURE REVIEW

2.1 Literature Review

The evolving landscape of e-commerce has been a subject of considerable scholarly attention, especially in the context of developing economies. Li and Zhang (2019) [2] emphasize the transformative impact of increased internet connectivity on e-commerce adoption, a trend reflected in Nepal's growing digital landscape. Chen, Li, and Sun (2020) [1] discuss the critical role of user experience (UX) in e-commerce success, underscoring the need for seamless transitions from traditional retail to online platforms. In the realm of security and privacy concerns, Wang, Zhang, and Zhang (2018) [5] provide valuable insights into encryption technologies and secure payment gateways, which are essential in building consumer trust in online transactions.

The infrastructure required for dedicated e-commerce platforms is explored by Liu and Li (2017) [3], who offer insights into the creation of efficient online marketplaces. This is especially relevant for Nepal, where a well-established e-commerce infrastructure for recreational goods like toys and sports materials can significantly impact economic growth. Molla and Licker (2005) [4] provide a theoretical framework on digital inclusion, highlighting the broader economic implications of e-commerce adoption in developing countries. Singh and Kapoor (2019) [6] further contribute by examining mobile commerce adoption, a critical component of e-commerce in urban environments.

In addition to these foundational studies, Johnson and Wong (2020) [7] explore ways to enhance user trust in e-commerce through improved privacy policies, while Zhao and Wang (2018) [8] examine e-commerce growth and consumer preferences in Asia, offering insights relevant to Nepal’s context. Chen and Liu (2021) [9] also investigate cross-border e-commerce adoption in developing markets, providing useful parallels for understanding Nepal’s positioning within the broader e-commerce landscape.

Collectively, these studies provide a comprehensive foundation for understanding the multifaceted challenges and opportunities in the integration of e-commerce, specifically for toys and sports goods in Nepal.

CHAPTER 3:
System Analysis and Design

3.1 System Analysis

Our learning website project follows an **incremental methodology**, which emphasizes iterative development, collaboration, and flexibility in responding to changing requirements and priorities. The system analysis phase is an integral part of the incremental development process, focusing on understanding user needs, defining project goals, and establishing a clear vision for the product.

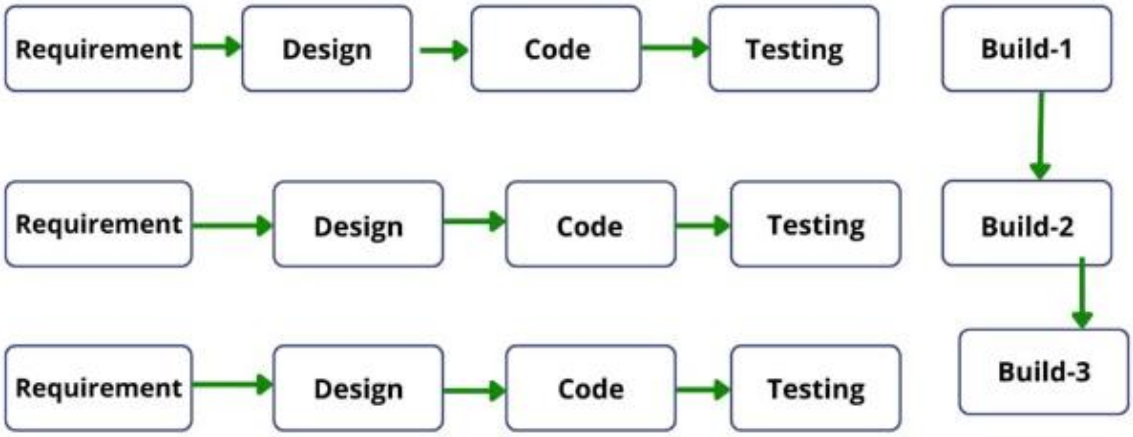


Figure 3.1 Incremental Methodology

We have chosen the incremental methodology for our project, following a cyclical and phased approach across multiple builds. Each build involves a complete cycle of requirement analysis, design, coding, and testing, resulting in progressively functional increments of the final product. This approach has been organized as follows:

1. **Requirement Analysis**: In each build cycle, requirements are gathered, refined, and prioritized through discussions with stakeholders, including end-users, product owners, and the development team. By using user stories, personas, and acceptance criteria, functional and non-functional requirements are captured in incremental phases. This allows for adjustments and refinements based on user feedback and evolving project needs.

2. **Design**: During the design phase for each build, high-level architectural designs and wireframes are created based on the identified requirements for that specific increment. The design is modular and flexible, allowing the addition of features in later builds. Collaborative decision-making within cross-functional teams ensures that design choices are aligned with both current and future needs.
3. **Implementation (Code)**: Incremental development involves breaking down the project into smaller, manageable components or features that are developed and delivered in stages. Each iteration results in a functional increment of the product. Development work is organized into short iterations, typically lasting 1-4 weeks, where developers implement and deliver these increments. Continuous integration and version control practices ensure that code changes are integrated and tested frequently.
4. **Testing**: Quality assurance and testing are integrated throughout the incremental development lifecycle. Automated testing tools and techniques are used to validate the functionality, performance, and usability of each product increment. Testing activities, including unit testing, integration testing, and user acceptance testing, are conducted collaboratively by developers, testers, and other stakeholders within each iteration.
5. **Maintenance**: Incremental projects prioritize responding to change and customer feedback over rigid adherence to a predefined plan. Therefore, maintenance activities, such as bug fixes, enhancements, and refinements, are incorporated into the ongoing development process. Continuous improvement and adaptation to evolving requirements and market conditions are central tenets of the incremental approach.

3.1.1 Requirement Analysis

Before beginning the project development process, it's crucial to gather all necessary requirements incrementally. This includes understanding what the system should do and how it should function. We've examined both the specific functionalities the system must provide (functional requirements) and the general qualities it should possess (non-functional requirements) as part of the incremental design and development process.

3.1.1.1 Functional Requirement

Different functional requirement of the system has been identified and are listed as below:

- **User Registration and Login**: The system should allow users to create accounts and log in with their credentials.
- **Products Viewing**: The system should display the available products and how much quantity is it available in.
- **Purchase Products**: Users should be able to purchase products that they desire.
- **Cancellation and Modification**: Admin should be able to cancel or modify their products listings and users in the system.
- **Admin Dashboard**: Administrators should have access to a dashboard to manage products, manage orders, manage users and resolve any issues along with being able to analyze different analytical tools available to them.

3.1.1.2 USE-CASE DIAGRAM

In the Website, there are two main actors: Admin and Users. The admin is responsible for managing the system and its components, while Users are individuals who interact with the system for commercial purposes like buying and ordering products.

The admin can log in to the system through an admin dashboard. Once logged in, the admin can manage various aspects of the system, including Users, Purchases and Product Listings. The admin can log out from the admin dashboard after completing tasks and activities.

Users, can register for an account within the system. After registration, Users can log in using their credentials.

After completing their tasks, Users can log out from the system.

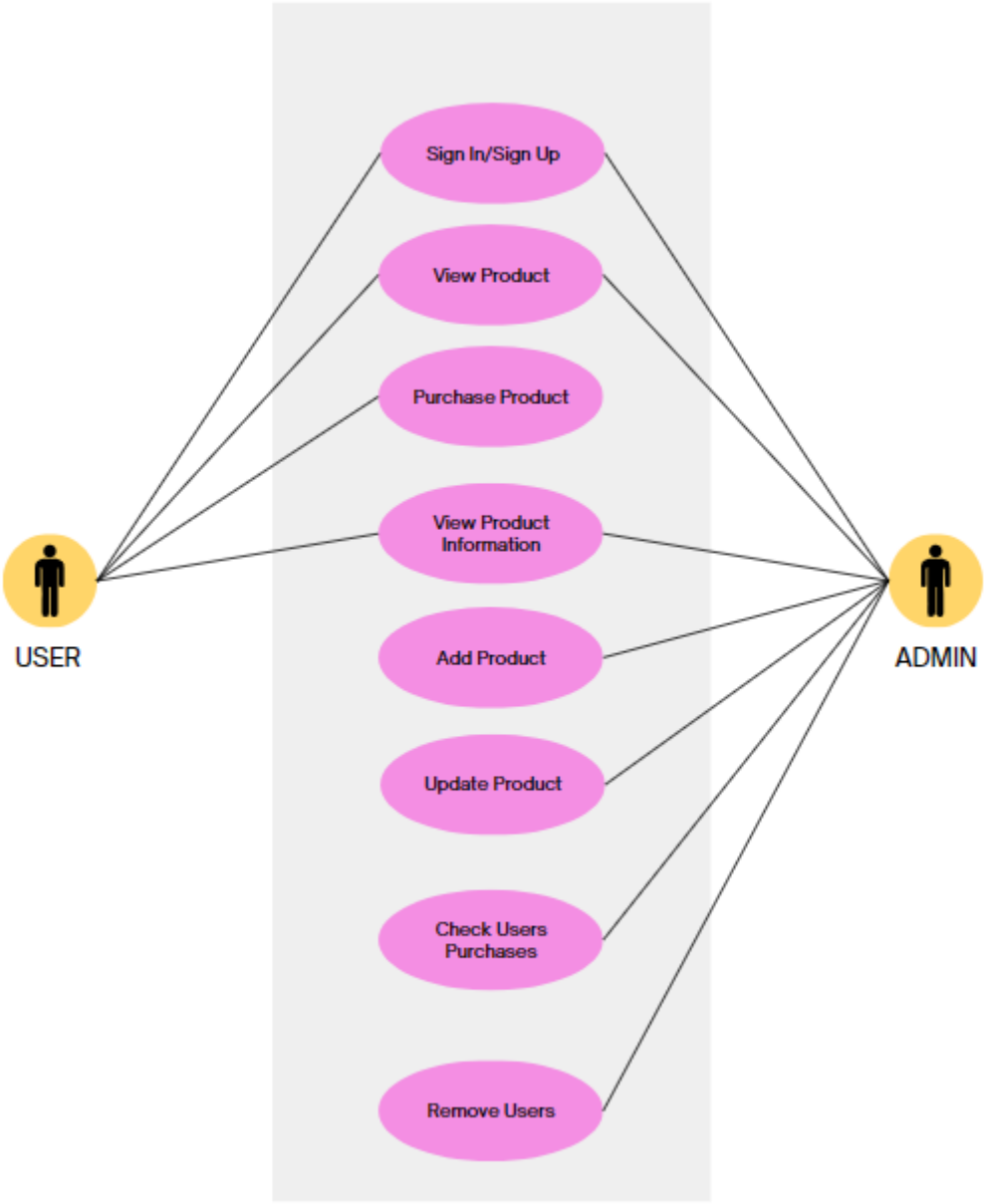


Figure 3.2 Use Case Diagram

3.1.1.3 Non-Functional Requirement

Here are the non-functional requirements for the Ecommerce Website:

- **User-Friendly Design:** The system is easy to use for both administrators and users. Moving around the system is simple and intuitive.
- **Security and Privacy:** User information are kept safe and private. Only authorized people have access to certain parts of the system and different validation process is used.
- **Performance**: - The performance of the system is fast and accurate as in this system database is normalized so it provides fast operations.

3.1.2 Feasibility Analysis

The project's feasibility study indicated that its implementation can be successful, thanks to its careful planning.

i. **Technical Feasibility:** The system's technical feasibility is evident, given that the needed resources for its development are readily obtainable. The necessary hardware and software are accessible, and the fundamental programming language is well-suited for the project. Moreover, the used libraries are capable of achieving the intended outcomes. Utilizing the existing resources for both development and ongoing maintenance of the system is feasible.
ii. **Economic Feasibility:** The system's economic feasibility is assured, being budget-friendly. All necessary tools and resources are either open source or available for free. Once the system is finished, the organization won't incur additional expenses for new hardware or software, since the existing resources can be utilized.
iii. **Operational Feasibility:** The system's operational feasibility is confirmed, as it aligns with practicality. The proposed tools and processes are manageable. The system's functions and features can be smoothly integrated into the existing operations. The potential users are familiar with similar systems, enhancing the system's acceptance and usability. Overall, the system's operational feasibility is promising, given its compatibility with the users' familiarity with similar solutions.
iv. **Schedule Feasibility Study:** The system is completed within scheduled time and do not exceed the scheduled time.

S.No	Task Name	Start Date	Finish Date
1	Getting Started & Planning	20/05/2024	06/06/2024
2	Analysis	06/06/2024	20/06/2024
3	Design	20/06/2024	30/07/2024
4	Coding	30/07/2024	15/08/2024
5	Testing	15/08/2024	24/08/2024
6	Documentation	20/05/2024	02/09/2024

Table 3.1 Gantt chart table



Figure 3.2 Gantt chart

3.1.3 Process Modeling (DFD)

Data Flow Diagram Ecommerce consists of two levels of DFD context diagram and level one DFD. Both these levels are used for making data flow diagram of ecommerce.

In context diagram, the login request, register, user information are the inputs where users and admin request for login, users register and login. The login response, details of users, the outputs where admin and user get response about login success or cancel, admin get details of users and users get logged in.

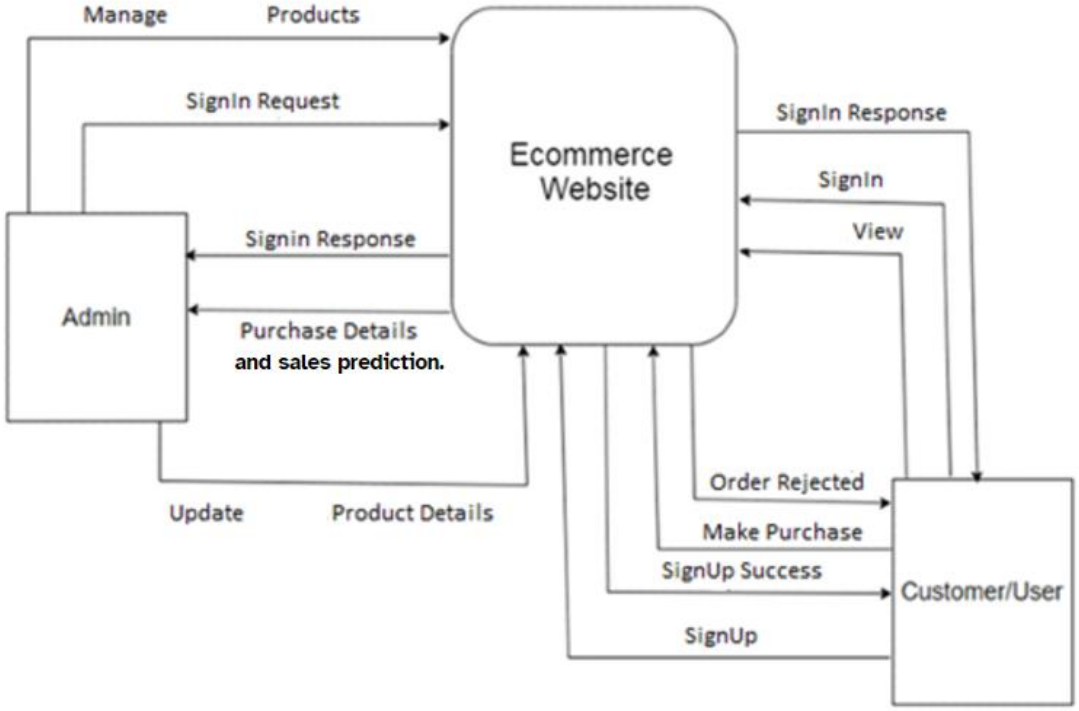


Figure 3.2 0 Level DFD Ecommerce Website

In level 0 DFD, This Data Flow Diagram (DFD) illustrates the interactions within an e-commerce website involving three main entities: Admin**,** e-commerce Website, and Customer/User**.** The Admin manages products by updating details, accessing purchase information, and requesting sign-in, with responses received from the website. The Customer/User can sign up, sign in, view products, and make purchases, with responses indicating success or failure for each action, such as a "Signup Success" or "Order Rejected" message. The e-commerce Website serves as the central hub, facilitating these

interactions by handling data flows related to product management, user authentication, and order processing, connecting both Admin and Customer/User actions seamlessly.

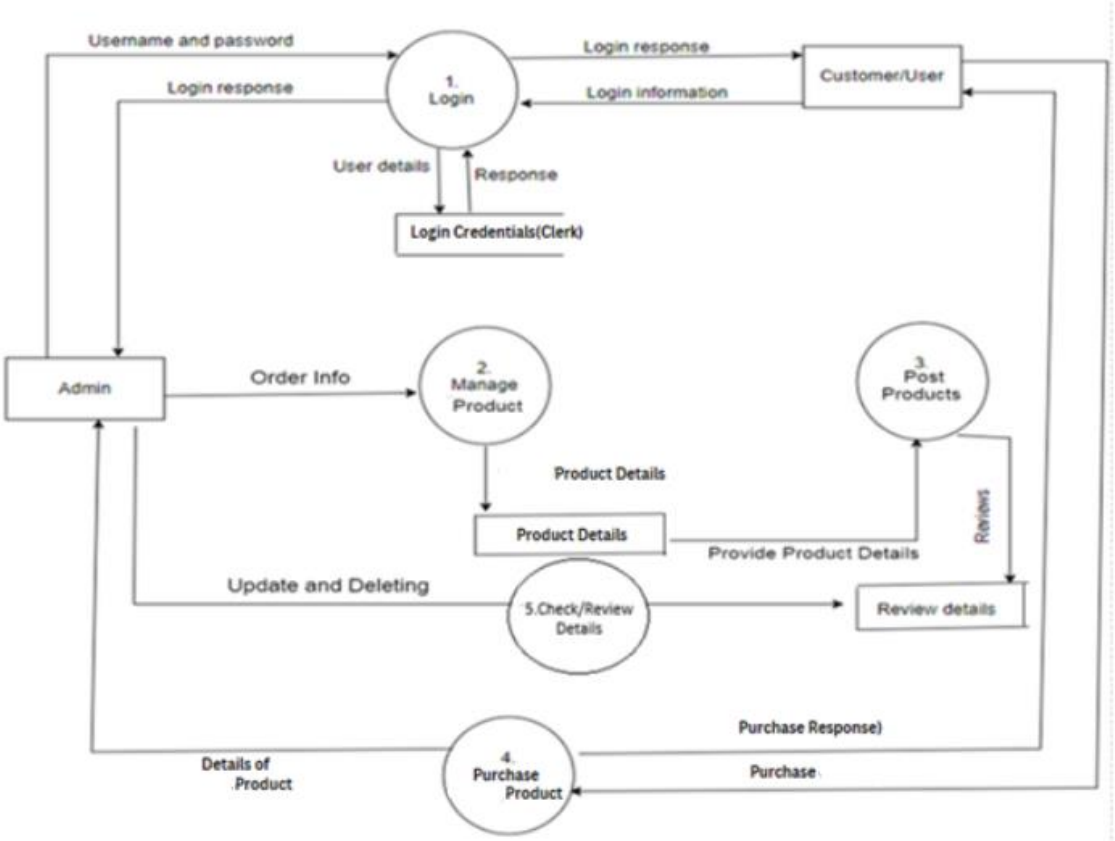


Figure 3.3 Level 1 DFD Ecommerce Website

In level 1 DFD, it illustrates key processes within an e-commerce platform, highlighting interactions between the Admin**,** Customer/User, and the system. It begins with Login, where both Admin and Customer/User authenticate with username and password, receiving a response from the system. The Admin can then manage products, updating or deleting product information through the Manage Product process. Products are posted by the system in the Post Products process, where details are provided upon request, allowing users to review them. The Customer/User can proceed to Purchase Product, receiving a purchase confirmation or response. Additionally, users can check and review product details before finalizing purchases. This DFD captures the flow of data across login, product management, posting, purchasing, and reviewing processes, ensuring that both Admin and Customer/User can effectively interact with the e-commerce platform.

3.2 System Design

To realize the different functional requirement of the system in graphical form, different design diagram of the system has been prepared which are as follows:

3.2.1 Architectural Design

This e-commerce platform's system is built on a modern architectural design, leveraging the Next.js framework for server-side rendering and enhanced performance. The architectural design comprises the following tiers:

- **Client Tier:**
    1. The client tier represents the user interface of the platform, where users interact with the application. It is responsible for rendering content and handling user interactions.
    2. Next.js, a React-based framework, is used to develop the client-side components of the application. It facilitates efficient client-side rendering and enables the creation of dynamic and interactive user interfaces.
    3. HTML, Tailwind CSS and Shadcn/UI are utilized to create the client-side content and implement the user interface elements. These technologies ensure a responsive and visually appealing user experience.
- **Web Tier**
    1. The web tier encompasses the server-side logic and application processing. It manages business logic, data persistence, and server-side rendering of the application.
    2. Next.js, as a hybrid framework, enables server-side rendering (SSR) and client-side rendering (CSR), providing enhanced performance and SEO benefits.
    3. Prisma, an ORM (Object-Relational Mapping) tool, is employed for database access and interaction. It simplifies database operations and ensures efficient data management.
- **Data Tier:**
    1. The data tier encompasses the database layer, which stores and manages the application's data. It provides persistent storage and retrieval of information required by the application.

2. CockroachDB, a NoSQL database, is utilized as the data store for the learning platform. CockroachDB's flexible document-based data model accommodates the dynamic nature of educational content and user data.
3. CockroachDB Atlas is employed as the cloud-hosted database service, offering scalability, reliability, and automated backups. It ensures seamless data management and high availability of resources for the application.
4. Overall, the architectural design of the learning platform leverages Next.js for efficient client-side and server-side rendering, Node.js for scalable backend development, Prisma for database interaction, and CockroachDB Atlas for reliable data storage and management. This three-tier architecture ensures robustness, scalability, and performance optimization for the e-commerce platform.

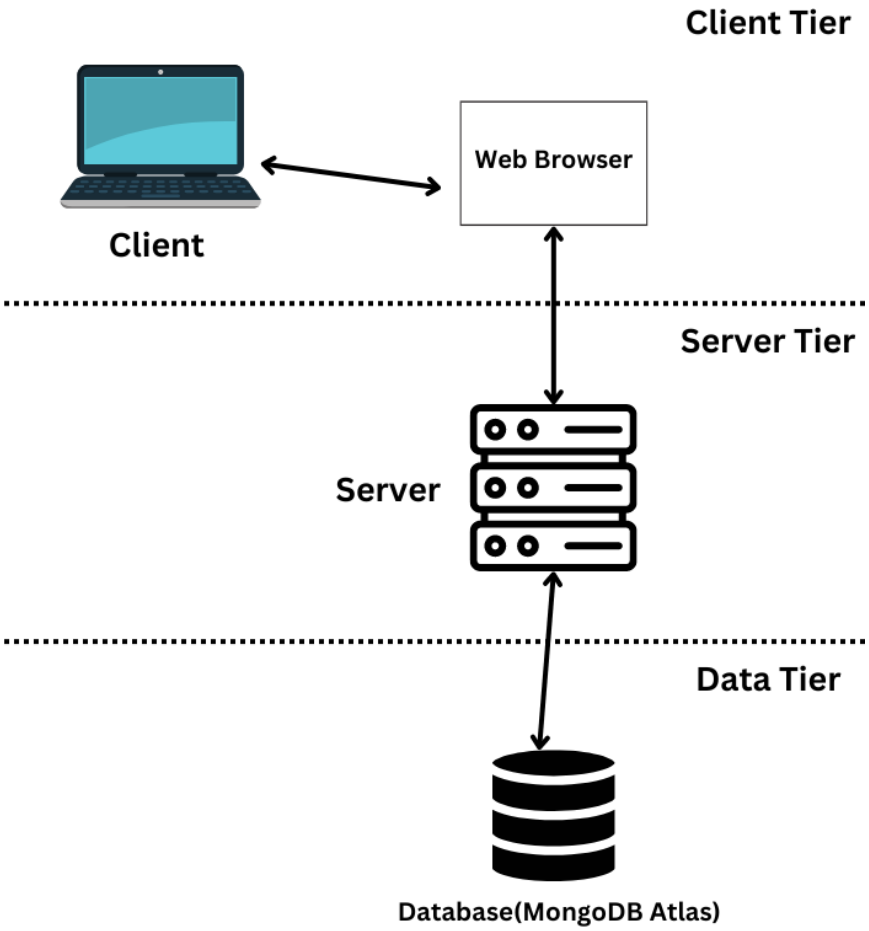


Figure 3.4 System Architectural design

3.2.2 System Flowchart

For Admin:

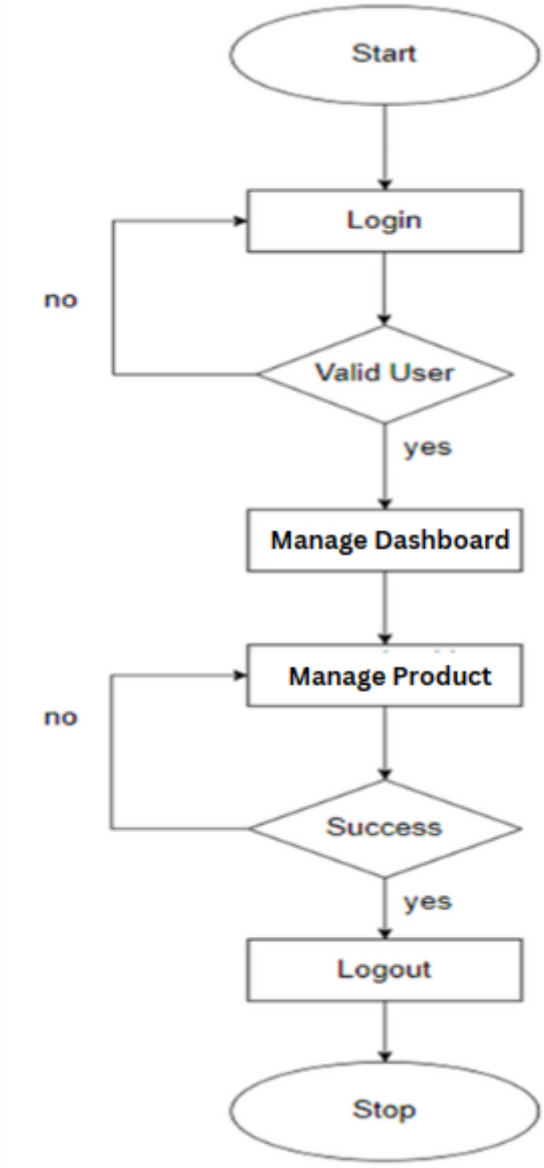


Figure 3.5 Admin Flowchart

Explanation for Admin Flowchart:

1**. Start**: The process begins with a starting point.

2. Admin Clicks "Login": This is the first action in the process, where the user initiates the login processes.

3. User Provides Credentials: The user enters their credentials provider i.e. Google.

4. System Validates Credentials: The system checks the provided credentials to see if they are valid.

5. Credentials Valid? A decision point to determine whether the entered credentials are valid or not.

- If the credentials are not valid (No), the system proceeds to display and redirect them to student dashboard.
- If the credentials are valid (Yes), the system grants access to the admin to the admin dashboard.

6. Admin Performs Actions: Once access is granted, the admin can perform various actions within the admin dashboard such as managing Products, purchases stats, etc.

7. Admin Clicks "Logout": When the admin decides to log out, they initiate the logout process.

8. End: The process ends, and the admin is logged out of the system.

For User:

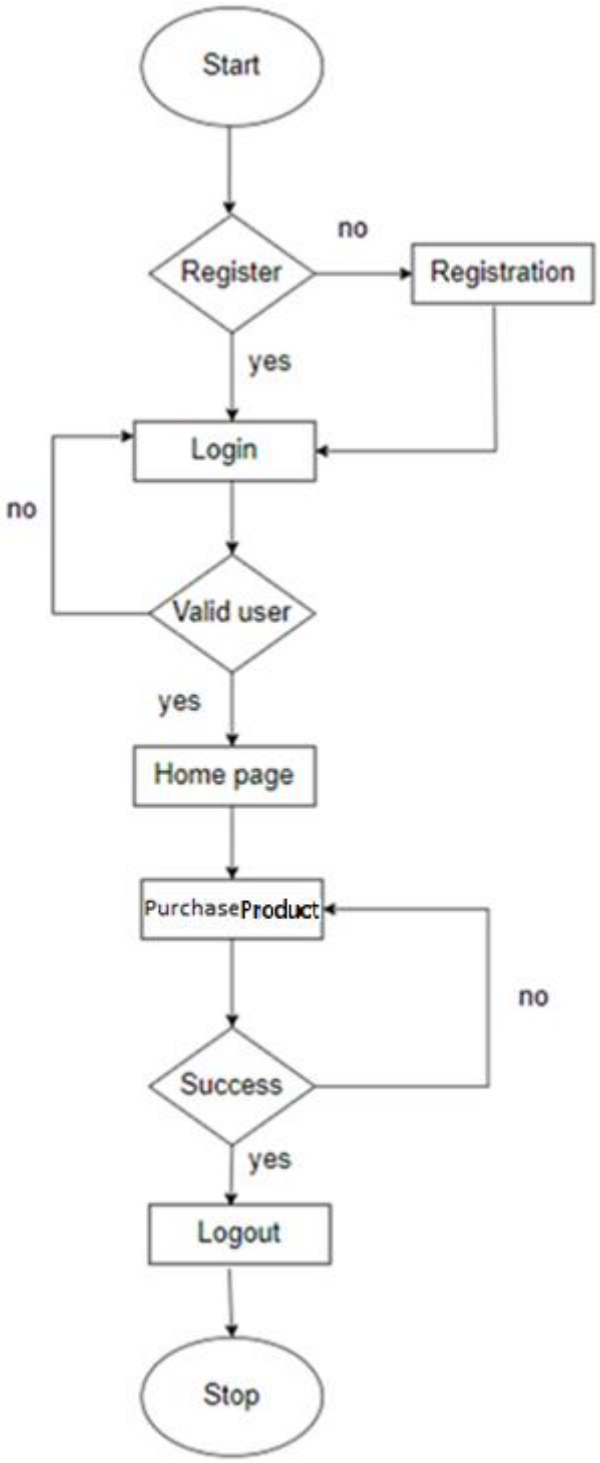


Figure 3.6 User Flow Chart

Explanation for User Flowchart:

1. **Start:** The process begins with a starting point.
2. **User Clicks "Sign in with Google":** The user initiates the authentication process by clicking on the "Sign in with Google" button.
3. **User is redirected to Google Login Page:** The user is redirected to the Google login page, where they can provide their Google credentials.
4. **User Provides Google Credentials:** The user enters their Google username and password.
5. **System Authenticates User with Google:** The system sends the user's provided Google credentials to the Google authentication service for verification.
6. **Authentication Successful** A decision point to determine whether the authentication with Google was successful.
7. **If the authentication is successful (Yes)**, the user is redirected back to the learning platform.
8. **If the authentication fails (No**), an error message is displayed, and the process is terminated.
9. **User is Redirected Back to the website**: Upon successful authentication with Google, the user is redirected back to the platform.
10. **User is Logged In:** The user is now logged into their account on the learning platform.
11. **User Navigates Through the Platform:** The user can now navigate through the platform, access Products, view their profile, and perform other actions.
12. **User Clicks "Logout":** If the user decides to log out, they initiate the logout process by clicking on the "Logout" button.
13. **End:** The process ends, and the user is logged out of their account on the learning platform.

3.2.3 Database Schema Design

The figure below is the database schema design of ecommerce website. Database schema design is used to show basic structure of the system. In ecommerce website, there are three tables in the databases each of them has their own fields where their id is primary key and if that id is used in another table, it becomes foreign key and foreign key are connected to another table with a line. There is data type of each entity and the foreign key in schema is represented by the arrow as shown in the diagram.

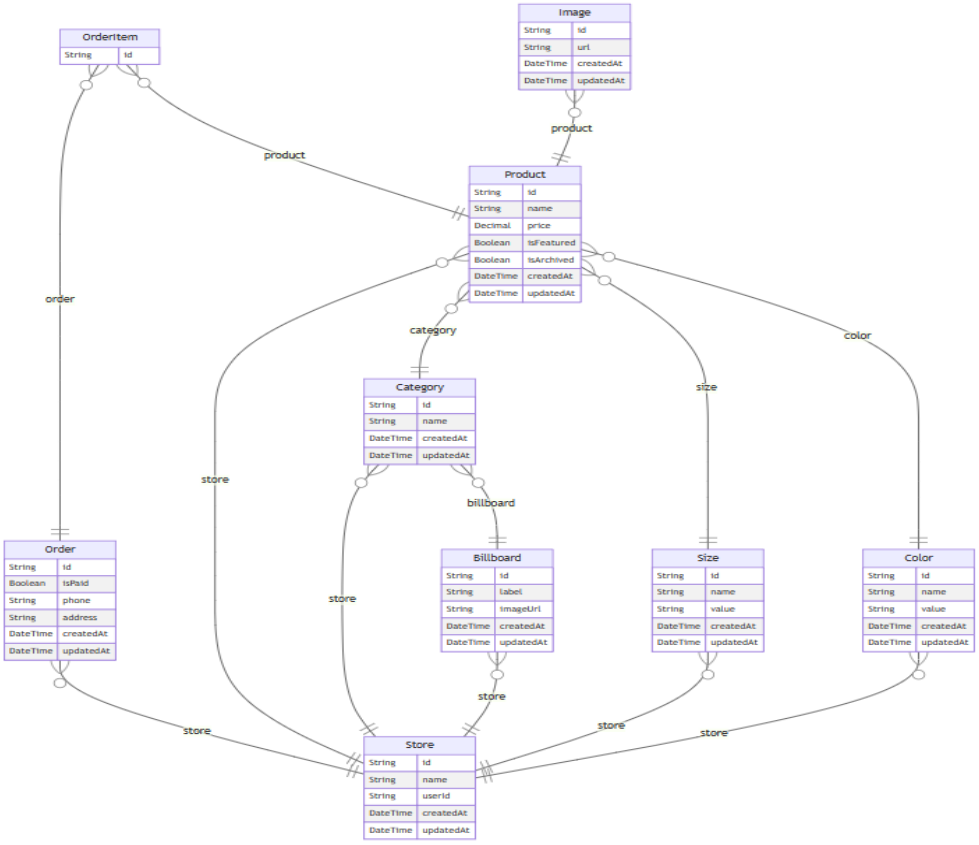


Figure 3.7 Database Schema

3.2.4 Interface Design

Before the real project design is completed, a few user interface designs are produced to demonstrate the user interface with the system while doing activities.



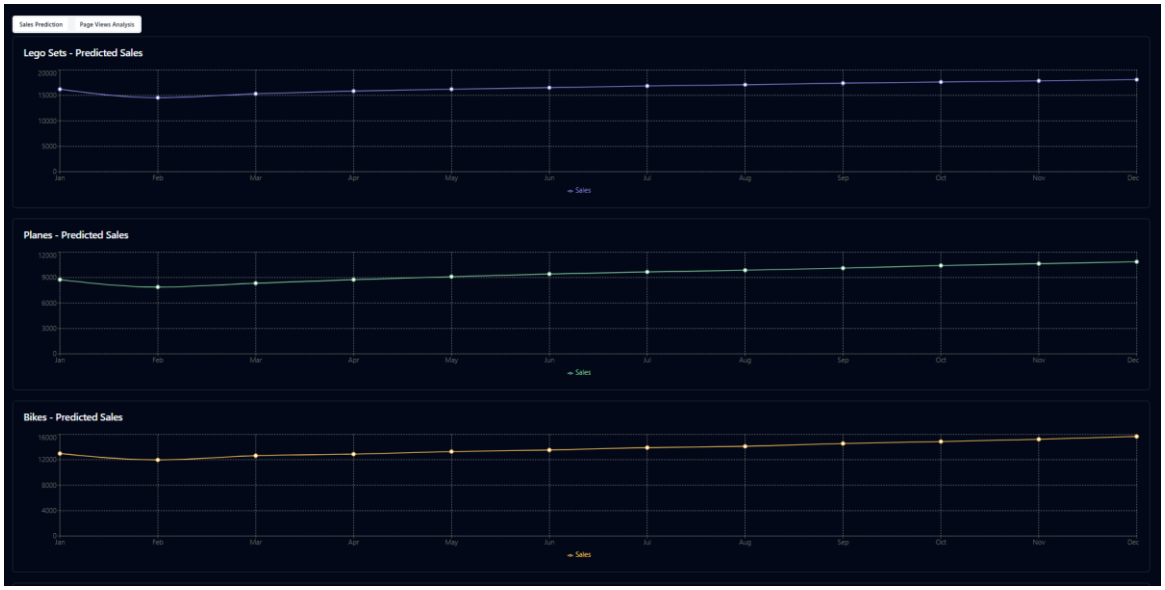


Figure 3.9 Admin Dashboard UI

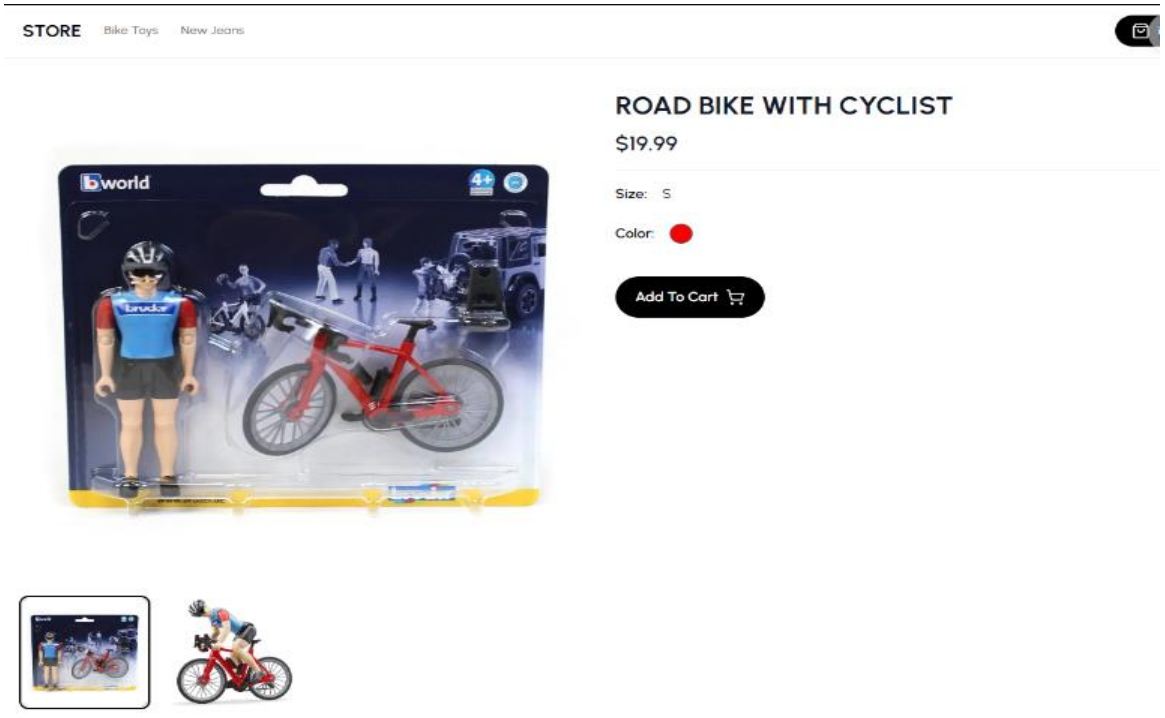


Figure 3.10 User Dashboard UI

CHAPTER 4:

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Utilized Tools (CASE tools, Programming language, Database platforms)

The implementation of the website utilizes a range of tools and frameworks to ensure efficient development and optimal user experience. Here's an overview of the tools and technologies employed in different layers of the application:

Front-End Tools:

- **Next.js with App Router:** Next.js is utilized as the primary framework for building the front-end of the learning platform. It provides server-side rendering, efficient routing with App Router, and a component-based architecture for building interactive user interfaces.
- **Tailwind CSS:** Tailwind CSS is employed for styling the components and layouts of the learning platform. It offers a utility-first approach to CSS, enabling rapid development and easy customization of styles.
- **Shadcn/UI Component Library:** Shadcn/UI provides a collection of pre-designed UI components that enhance the visual appeal and functionality of the learning platform. These components include buttons, forms, navigation menus, and more.
- **TypeScript:** TypeScript is used to add static typing to JavaScript code, improving code quality, and catching errors at compile-time. It enhances developer productivity and ensures better code maintainability.

Back-End Tools:

- **Clerk for Authentications:** Clerk is integrated into the back-end of the learning platform to handle user authentication using Google provider. It provides secure authentication flows, user management, and access control functionalities.
- **Next.js (Server-side):** It provides server-side rendering, efficient routing with App Router, and a component-based architecture for building interactive user interfaces.
- **Prisma as ORM:** Prisma serves as the Object-Relational Mapping (ORM) tool for the learning platform, enabling seamless interaction with the CockroachDB Atlas database. It simplifies database operations, data modeling, and schema migrations.

- **Stripe for Payments:** Stripe is integrated into the back-end of the learning platform to facilitate online payments for Product purchases. It offers a secure and flexible payment processing solution, supporting various payment methods and currencies.

Database:

- **CockroachDB Atlas:** CockroachDB Atlas is utilized as the cloud-hosted NoSQL database service for storing and managing data related to Products, users, transactions, and other application entities. It provides scalability, reliability, and automated backups, ensuring seamless data management for the platform.

Documentation Tools

- **MS Office:** This is used for writing and editing the documentation of the e-commerce system.
- **Draw.io:** This is used to generate diagrams for system analysis and design of e-commerce website "Phenomenol toys". Diagrams were created using this tool in order to save time since all components are available with drag and drop functions.

4.1.2 Implementation Details of Modules (Description of procedures/functions)

Different modules of this system are described as below:

Admin Module

- **Manage Users:** Admin can view a list of all registered users through clerk. Admin can search, filter, and sort user's name, or registration date. Admin can edit user or remove accounts as needed. Admin can view the predicted sales in the dashboard in the form of graphical representation.

View List of Products

- Admin has access to a user-friendly list view of all Products.
- Each product listing should display key information like name and status.
- Admin can quickly identify purchased Products.

View Purchases:

- Admin can view a list of all purchases made by users.
- Admin can search, filter, and sort orders.

View List of Purchases Stats:

- Admin can access a comprehensive list of orders for better oversight.
- Admin can access check the number of products and number of times a single product is purchased by the user/users.

View sales prediction of products:

- Admin can access a graphical representation of sales prediction for better planning.
- Admin can check the number of products that may be required and number of times a single product could be purchased by the user/users in the future.

User Module

Users can create new accounts and sign in with clerk after being authenticated by their providers. Registered users can log in securely using their email and password. User authentication and session management are crucial for securing user data and ensuring a seamless user experience. Users can view a list of available products for order. Users can click on a Product to view more detailed information, such as images and description. Users can select a Product and proceed to make a purchase through stripe. Upon successful purchase, users is scheduled for the delivery of the product.

4.2 Testing

System testing is done by giving different training and testing datasets. This test is done to evaluate whether the system is providing accurate summary or not. During the phase of the development of the system, our system is tested time and again. The series of testing conducted are as follow:

4.2.1 Test Cases for Unit Testing

During unit testing, a modularized approach was employed, with individual modules being thoroughly tested. The system concentrated on each module until accurate results were achieved. Input forms underwent testing to ensure they rejected invalid inputs.

Table 4.2.1: User Sign in/Sign up Test Case

S. N	Preconditions	Actions	Expected Outcome	Actual Outcome
Testing for User Authentication				
1.	User has an existing Google account. User navigates to the learning platform's login page.	• User clicks on the "Sign in with Google" button.	User is redirected to the Google authentication page. User provides their Google credentials and authenticates successfully. User is redirected back to the website and logged in automatically. User gains access to their account dashboard and other authenticated features.	User is successfully redirected to the Google authentication page. User provides Google credentials and authenticates. User is redirected back to the platform. User gains access to their account dashboard and authenticated features.

4.2.2 Test Case for System Testing

In system testing, whole system is tested as below:

Table 4.2.2: System Test Case

S. N	Preconditions	Actions	Expected Outcome	Actual Outcome
Testing for Product Searching				
1.	User is logged into their account on the platform. User navigates to the search bar or search page.	• User enters keywords or filters to search for a specific Product. • User clicks on search results or scrolls through the Product catalog.	Relevant Products matching the search criteria are displayed. Each Product listing includes essential information such as title, description, and price. User can easily identify and select desired Products from the search results.	Search functionality retrieves relevant Products based on keywords or filters. Product listings contain detailed information. User can easily navigate through search results and select desired Products.

S. N	Preconditions	Actions	Expected Outcome	Actual Outcome
Testing for Product Viewing				
2.	User navigates to the searches for a specific Product.	• User clicks on a specific Product to view detailed information.	Detailed Product information is displayed, including title, description, and pricing. User can navigate through Product details and review content.	User successfully views detailed information about the selected Product. Product details are comprehensive and accurately represent the Product content.

Testing for Stripe Payment Integration				
3.	User has selected a Product to enroll in and proceeded to the checkout page.	• User enters payment information, including credit card details. User clicks on the "Pay Now" or "Complete Purchase" button to initiate payment.	Payment processing is initiated securely through the integrated Stripe payment gateway. User's credit card information is securely transmitted and validated. Upon successful payment processing, user receives a payment confirmation message.	Payment processing is initiated securely through Stripe. User's credit card information is securely transmitted and validated. Payment confirmation message is displayed upon successful processing.

4.2.3 Test Case for Admin

Table 4.2.3: Admin Dashboard Test Case

S. N	Preconditions	Actions	Expected Outcome	Actual Outcome
Admin Login				
1.	The admin is on the login page.	• User clicks on the "Sign in with Google" button.	The admin should be successfully logged in and redirected to the admin dashboard.	The admin should be successfully logged in and redirected to the admin dashboard.
Adding a New Product				
2.	The admin is logged in and on the admin dashboard.	• Admin clicks on the "Add New Product" button. • Admin fills in the required information for the new Product, including title, description, chapters, and price. • Admin uploads Product materials and resources. • Admin selects whether to publish or unpublish the Product.	New Product is successfully added to the platform. Product details are accurately captured, including title, description, and price.	New Product is successfully added to the platform. Product details are accurately captured, including title, description, and price.
Editing an Existing Product:				
3.	The admin is on the orders list page.	• Admin navigates to the Product management section.	Changes to the Product details are successfully saved. Updated	Changes to the Product details are successfully saved. Updated

		• Admin selects the Product they want to edit. • Admin makes changes to the Product details, such as updating the description, or adjusting the price.	Product information is reflected accurately on the platform. Users can view the updated Product details.	Product information is reflected accurately on the platform. Users can view the updated Product details.
Publishing/Unpublishing a Product:				
4.	The admin is on the orders list page.	• Admin navigates to the Product management section. • Admin selects the Product they want to publish or unpublish. • Admin toggles the publish/unpublish switch to the desired state.	Product is published or unpublished based on the admin's selection. Unpublished Products are hidden from users and not accessible until republished by the admin.	Product is published or unpublished based on the admin's selection. Unpublished Products are hidden from users and not accessible until republished by the admin.

Analyzing the sales :				
5.	The admin is logged in and on the Orders List page. Predicted sales data is available.	• Admin navigates to the Sales Analysis or Reporting section. • Admin views available sales metrics, such as sales volume, revenue trends, and	The admin is provided with comprehensive sales data prediction for the selected product.	Sales data is displayed accurately and relevant metrics like total sales volume, revenue trends, and sales peaks

		time-based sales insights.		are presented. Data visualizations (charts, graphs) are populated correctly, providing clear insights into sales performance for the selected product.

CHAPTER 5:

CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Knowledge Gained / Results

Every project offers a platform for learning and acquiring knowledge across various dimensions. In this particular project, we gained valuable problem-solving abilities and honed independent problem-solving, adhered to guidelines effectively, improved communication and writing proficiencies, and practiced team management.

- **Problem-Solving Skills**

This project significantly enhanced our problem-solving skills, equipping us to identify and troubleshoot various system-related issues.

- **Enhanced Writing Abilities**

Our writing skills improved as we learned to craft proposals and project documentation. Additionally, we became proficient in utilizing diverse case tools for diagrams like use case, schema, data flow, and ER diagrams.

- **Time Management**

A crucial takeaway was the skill of managing time according to the complexity of system components. This enabled us to prioritize tasks effectively.

- **Optimize User Experience and Transition**

Smooth transition statistics, including reduced bounce rates and increased time spent on the website.

5.2 Conclusion

This ecommerce website aims to give a smooth and user friendly experience for both the user and admin where they can order products effectively and seamlessly. It also aims to encompass the essence of ecommerce.

5.3 Future Recommendations

The development project could have been more efficiently handled with regards to design and development. The documentation process might have been better programming the project prior to any documentation. The system can be updated based on the users' requirements recommendation. The page load and server load speed might be improved.

Some of the future recommendation for this system are:

- **User Feedback:** Continuously refine features and offerings based on customer input.
- **Machine Learning:** Implement algorithms for personalized shopping experiences.
- **Expand Offerings:** Diversify the range of toys and related products available.
- **Mobile Optimization:** Ensure the platform is responsive across all devices.
- **Community Features:** Introduce forums for parents and enthusiasts to share reviews and recommendations.
- **Security Enhancements:** Regularly update security protocols to protect customer data.
- **Partnerships:** Collaborate with toy manufacturers and educational institutions to broaden product offerings and reach.

REFERENCES

[1] Y. Chen, X. Li, and M. Sun, "Enhancing user experience in e-commerce: A review," Int. J. Hum.-Comput. Interact. vol. 36, no. 7, pp. 669–682, 2020.

[2] X. Li and L. Zhang, "The impact of internet connectivity on e-commerce adoption: Evidence from rural China," Inf. Technol. Dev., vol. 25, no. 4, pp. 599–619, 2019.

[3] C. Liu and X. Li, "Building efficient online marketplaces: A review," Int. J. Electron. Commer., vol. 21, no. 4, pp. 508–530, 2017.

[4] Molla and P. S. Licker, "E-commerce adoption in developing countries: A model and instrument," Inf. Manage., vol. 42, no. 6, pp. 877–899, 2005.

[5] X. Wang, L. Zhang, and L. Zhang, "Secure e-commerce transactions: A comprehensive review," Inf. Syst. Front., vol. 20, no. 5, pp. 1057–1075, 2018.

[6] Singh and R. Kapoor, "Examining mobile commerce adoption in urban India," J. Emerg. Mark. E-Commer., vol. 34, no. 3, pp. 450–467, 2019.

[7] M. Johnson and K. Wong, "Enhancing user trust in e-commerce through improved privacy policies," J. E-Commer. Secur., vol. 19, no. 4, pp. 377–390, 2020.

[8] H. Zhao and Y. Wang, "E-commerce growth and consumer preferences in Asia," J. Global Bus., vol. 22, no. 6, pp. 660–678, 2018.

[9] J. Chen and Y. Liu, "Understanding cross-border e-commerce adoption in developing markets," J. Int. E-Commer., vol. 32, no. 1, pp. 25–43, 2021.